states of two controlled qubits.

$$\text{CCX}=\begin{bmatrix} 1&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 1&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0 \end{bmatrix}$$

The following piece of code applies a Toffoli gate (Figure 9):

```
qc = QuantumCircuit(3)
qc.ccx(0,1,2)
qc.draw(output='mpl')
```

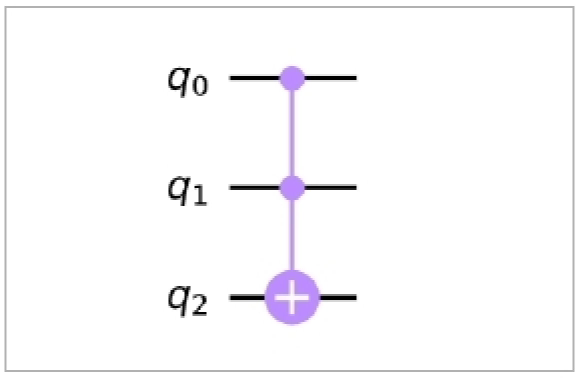

Figure 9: A CCNOT (CCX) or a Toffoli gate

There are many gates provided by the Qiskit circuit library. We have mentioned a few very important and basic gates that one should be familiar with before diving into quantum computing. Now that we have learned about these various gates, we can use them to create different types of quantum phenomena, two of which are 'superposition' and 'entanglement'. We will now show you how to experience this in a practical way.

Superposition

We mentioned earlier that the Hadamard gate is known to put any qubit in an equal superposition of two states. In this state, the qubit can be observed to have equal amplitudes of both states.

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

This means that a measurement of this qubit at any point will have equal probabilities of being $|1\rangle$ or $|0\rangle$, creating a 'superposition' of states.

How to find the probabilities:

Probability of the resultant state $|\psi\rangle$ being equal to $|0\rangle$ can be found by getting the inner product of both:

$$\langle\psi\rangle = \frac{1}{\sqrt{2}}\langle 0\rangle + \frac{1}{\sqrt{2}}\langle 1\rangle$$

We know that in case of inner products, $\langle 0|0\rangle = 1$ and $\langle 0|1\rangle = 0$; so:

$$\langle 0|\psi\rangle = \frac{1}{\sqrt{2}}$$

$$|\langle\psi\rangle|^2 = \frac{1}{2}$$

$$= 0.5$$

Similarly, calculate the probability of the resultant state $|\ \rangle$ being equal to $|1\rangle$:

$$\langle\psi\rangle = \frac{1}{\sqrt{2}}\langle 0\rangle + \frac{1}{\sqrt{2}}\langle 1\rangle$$

$$= \frac{1}{\sqrt{2}}$$

$$|\langle\psi\rangle|^2 = \frac{1}{2}$$

$$= 0.5$$

Now, let's check that programmatically:

```
qc = QuantumCircuit(1)
qc.h(0)
qc.draw(output='mpl')
```

Let's plot a histogram to find out the probabilities of both states occurring after execution:

```
from qiskit.visualization import plot_
histogram
backend = Aer.get_backend('statevector_
simulator')
results = execute(qc,backend).result().
get_counts()
print(results)
plot_histogram(results,
color='darkturquoise')

Results: {'0': 0.5, '1': 0.5}
```

This shows that there is equal probability for 'q' to be either in the $|0\rangle$ state or in the $|1\rangle$ state. Figure 10 shows the circuit output and the plotted histogram for probabilities.

Entanglement

Quantum entanglement is a quantum mechanical phenomenon in which the quantum states of two or more objects have to be described with reference to each other, even though the individual objects may be spatially separated. This leads to correlations between observable physical properties of the systems. We mentioned while explaining the CNOT gate that it helps in establishing the entanglement phenomena. With the following circuit you can observe entanglement.

```
qc=QuantumCircuit(2)
qc.h(0)  # Hadamard gate on q0
qc.cx(0,1)  # CNOT on q1 controlled on
q0
qc.draw('mpl')
```

As the H gate is known to put any qubit in equal superposition of $|0\rangle$ and

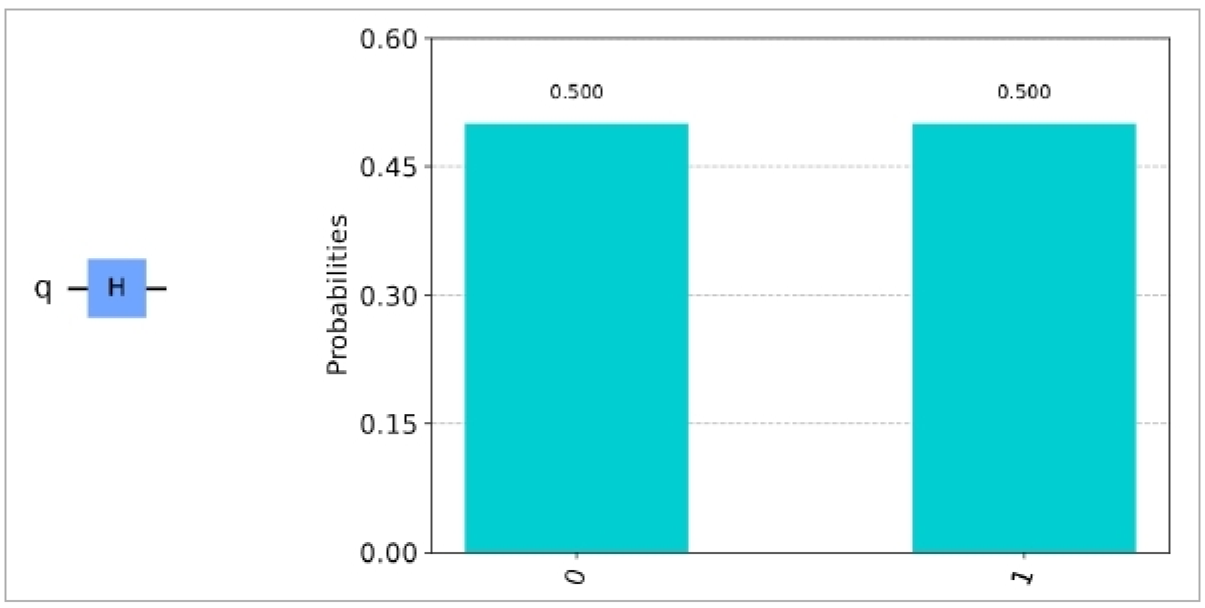

Figure 10: The H gate applied on a single qubit places it in an 'equal superposition' of two states 0 and 1